

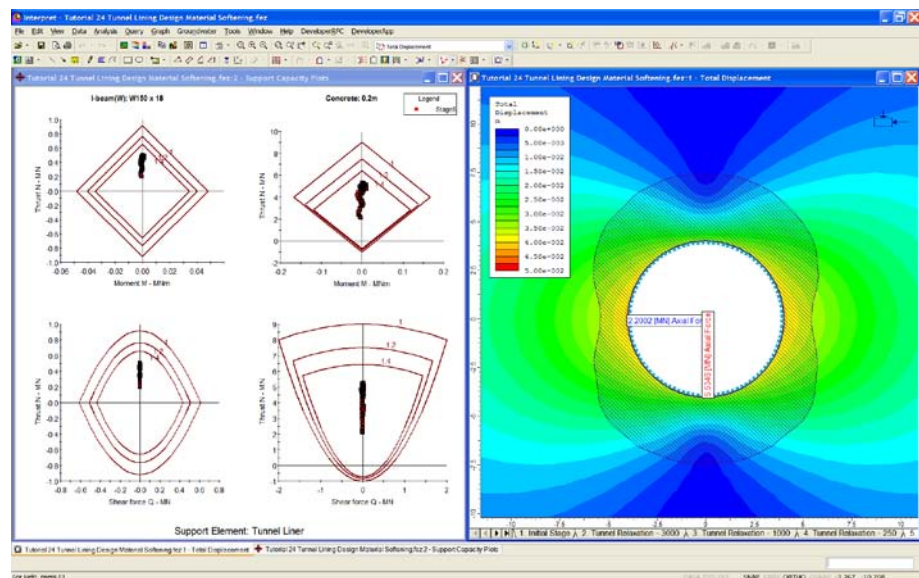
# 3D Tunnel Simulation using the Core Replacement Technique

In this tutorial, *Phase2* is used to simulate the three-dimensional excavation of a tunnel. In three dimensions, the tunnel face provides support. As the tunnel face advances away from the area of interest, the support decreases until the stresses can be accurately modelled with a two-dimensional plane-strain approach. This procedure is necessary in order to determine the amount of deformation prior to support installation.

The complete model can be found in the **Tutorial 18 3D Tunnel Simulation using Core Replacement.fez** file, which can be accessed by selecting File > Recent Folders > Tutorials Folder from the *Phase2* main menu.

## Topics covered

- 3D tunnel simulation
- Core Replacement Technique (Material Softening)
- Reinforced concrete liners
- Support capacity curves



## Problem

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A circular tunnel of radius 4m is to be constructed in Schist at a depth of 550m. The in-situ stress field has been measured with the major in-plane principal stress equal to 30 MPa, the minor in-plane principal stress equal to 15 MPa and the out-of-plane stress equal to 25 MPa. The major principal stress is horizontal and the minor principal stress is vertical. The strength of the Schist can be represented by the Generalized Hoek-Brown failure criterion with the uniaxial compressive strength of the intact rock equal to 50 MPa, the GSI equal to 50 and  $m_i$  equal to 10. To compute the rock mass deformation modulus, the modulus ratio (MR) is assumed to be 400. The support is to be installed 2m from the tunnel face.

The goal of this tutorial is to demonstrate how to model the tunnel deformation prior to support installation using the core replacement (material softening) approach.

To design a support system, the following procedure can be used:

1. Determine the amount of tunnel wall deformation prior to support installation. As a tunnel is excavated, there is a certain amount of deformation, usually 35-45% of the final tunnel wall deformation, before the support can be installed. Determining this deformation can be done using either a) observed field values, or b) numerically from 3D finite-element models or axisymmetric finite-element models, or c) by using empirical relationships such as those proposed by Panet or Vlachopoulos and Diederichs.
2. Using the core replacement technique, determine the modulus reduction sequence that yields the amount of tunnel wall deformation at the point of and prior to support installation. This is the value determined in step 1.
3. Build a model that relaxes the boundary to the calculated amount in step 2. Add the support and determine whether a) the tunnel is stable, b) the tunnel wall deformation meets the specified requirements, and c) the tunnel lining meets certain factor of safety requirements. If any of these conditions are not met, choose a different support system and run the analysis again.

## Model

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The first step is to determine the amount of tunnel wall deformation prior to support installation. For this tutorial, we will use the relationship proposed by Vlachopoulos and Diederichs. The Vlachopoulos and Diederichs method is documented in Appendix 1 of the Kersten Lecture by Hoek, Carranza-Torres, Diederichs and Corkum. The paper is in the Hoek's published papers area on the Rocscience website:

<http://www.rocscience.com/hoek/references/Published-Papers.htm>

This method requires that we build a model of the tunnel and determine a) the deformation far from the tunnel face using a simple plane strain analysis, and b) for the same model determine the plastic zone radius.

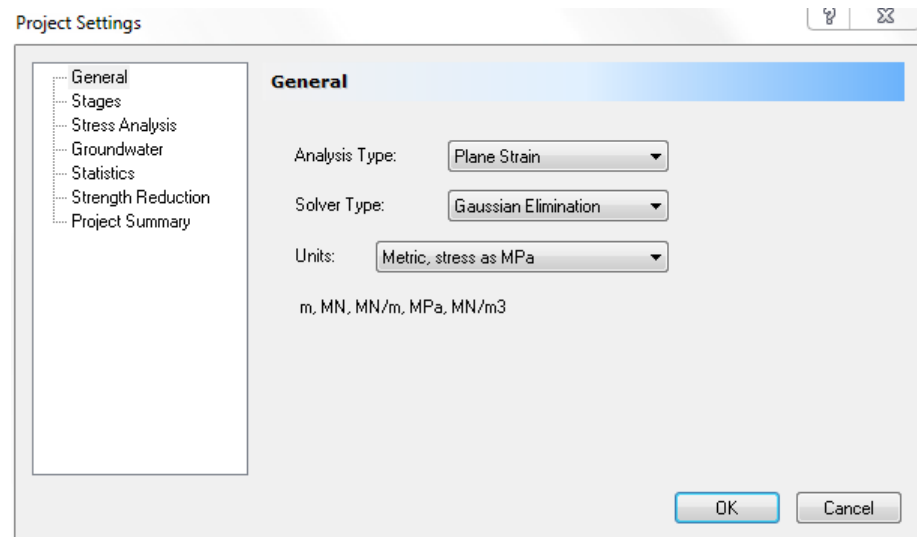
In this tutorial we'll start by building a single model that also combines step 2 with step 1. We'll build a plane strain model that sequentially replaces and reduces the modulus of the material inside the excavation over a number of stages. The final stage, with the material excavated inside the tunnel, will be used to determine the amount of deformation prior to support installation (step 1). The factoring of the modulus over a number of stages will be used to determine the modulus that yields the amount of tunnel wall deformation at the point of support installation (step 2).

Start the *Phase2* Model program.

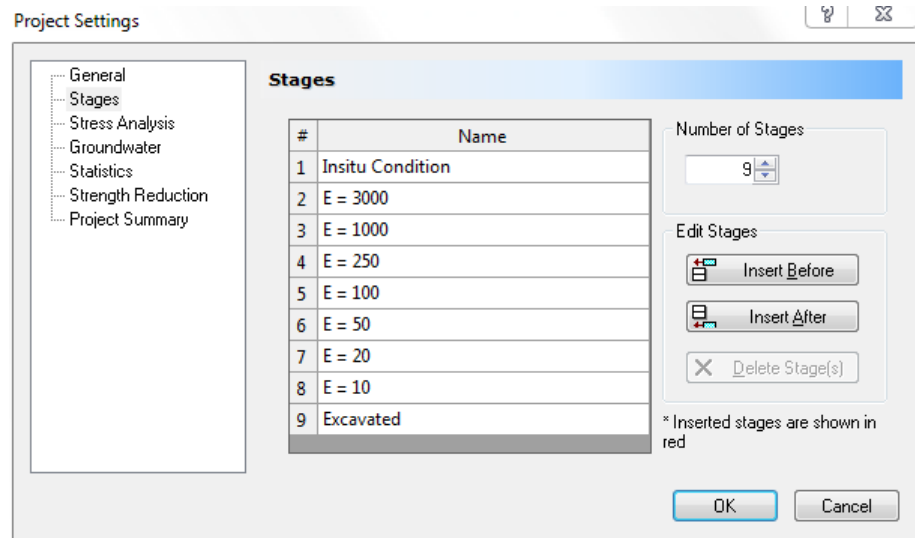
## Project Settings



Open the **Project Settings** dialog from the **Analysis** menu and make sure the General tab is selected. Define the units as being “Metric, stress as MPa”.



Select the Stages tab. Change the number of stages to 9 (see following figure). Fill in the stage names as seen below. Close the dialog by clicking OK.



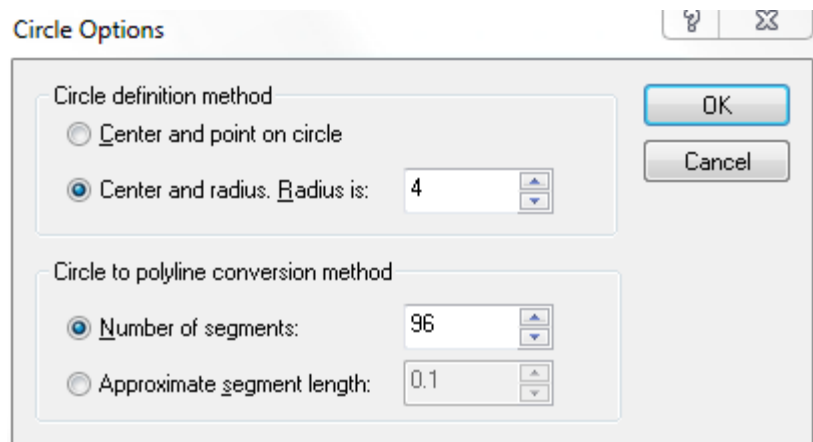
## Geometry

Now enter the circular tunnel.



Select: Boundaries → Add Excavation

1. Right-click the mouse and select the Circle option from the popup menu. You will see the following dialog.



2. Select the Center and radius option, enter Radius = 4 and enter Number of Segments = 96 and select OK.
3. You will be prompted to enter the circle center. Enter 0,0 in the prompt line, and the circular excavation will be created.



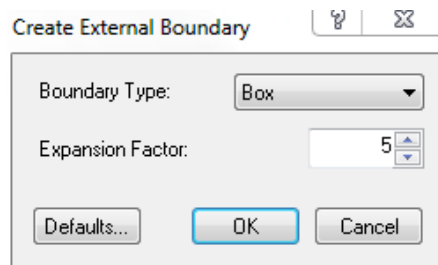
Select Zoom All (or press the F2 function key) to zoom the excavation to the center of the view.

Now we will create the external boundary. In *Phase2*, the external boundary may be automatically generated, or user-defined. We will use one of the 'automatic' options.



Select: Boundaries → Add External

You will see the Create External Boundary dialog. We will use the settings of Boundary Type = Box and Expansion Factor = 5. Select OK, and the external boundary will be automatically created.

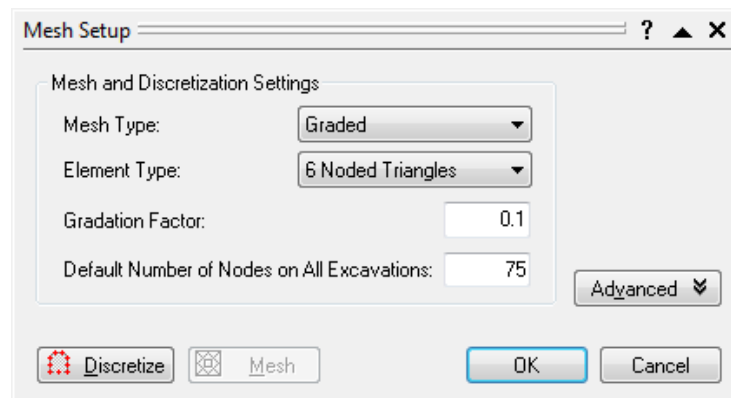


The boundaries for this model have now been entered.

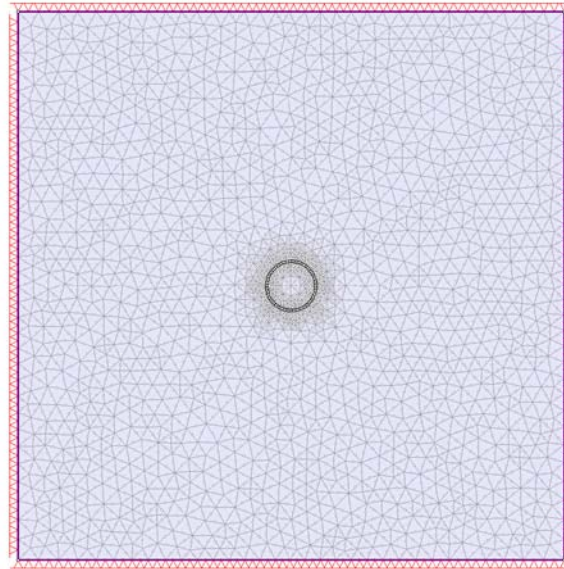
## Mesh



Add the finite element mesh by selecting **Mesh Setup** from the **Mesh** menu. In the mesh setup dialog, change the Element Type to 6 Noded Triangles.



Click the Discretize button and then the Mesh button. Click OK to close the dialog. The mesh will look like this:



## Boundary Conditions

For this tutorial, no boundary conditions need to be specified by the user. The default boundary condition will therefore be in effect, which is a fixed (i.e. zero displacement) condition for the external boundary.

## Field Stress

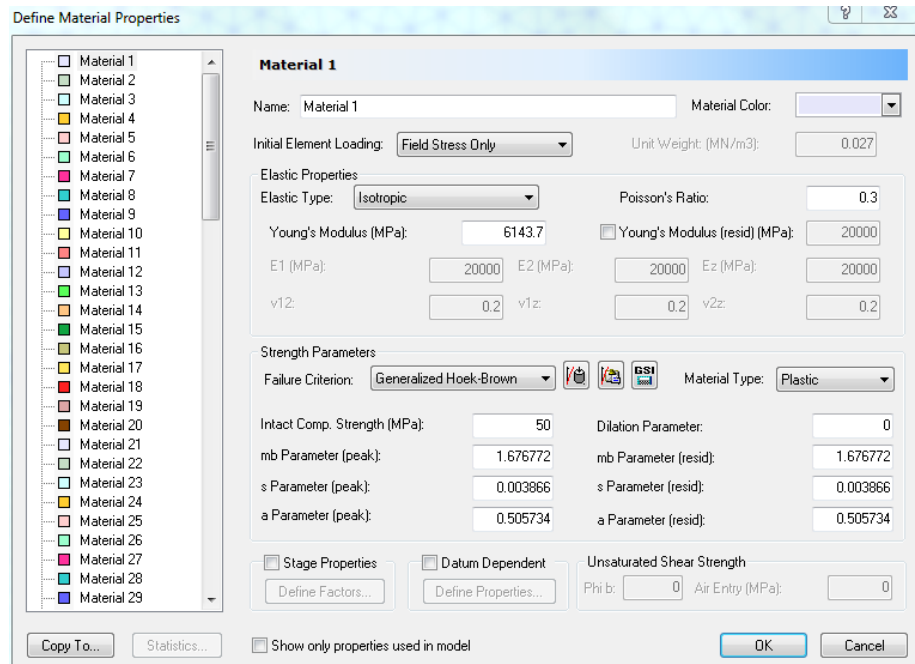
Field Stress determines the initial in-situ stress conditions, prior to excavation. As described earlier in this tutorial, the in-situ stress field has been measured with the major in-plane principal stress equal to 30 MPa, the minor in-plane principal stress equal to 15 MPa and the out-of-plane stress equal to 25 MPa. The major principal stress is horizontal and the minor principal stress is vertical.

Select: Loading → Field Stress

| Field Stress Type:                                         | Value |
|------------------------------------------------------------|-------|
| Constant                                                   |       |
| Sigma 1 (MPa, Comp. +):                                    | 30    |
| Sigma 3 (MPa, Comp. +):                                    | 15    |
| Sigma Z (MPa, Comp. +):                                    | 25    |
| Angle (degrees from horizontal, CCW):                      | 0     |
| Locked-in horizontal stress (in plane) (MPa, Comp. +):     | 0     |
| Locked-in horizontal stress (out-of-plane) (MPa, Comp. +): | 0     |

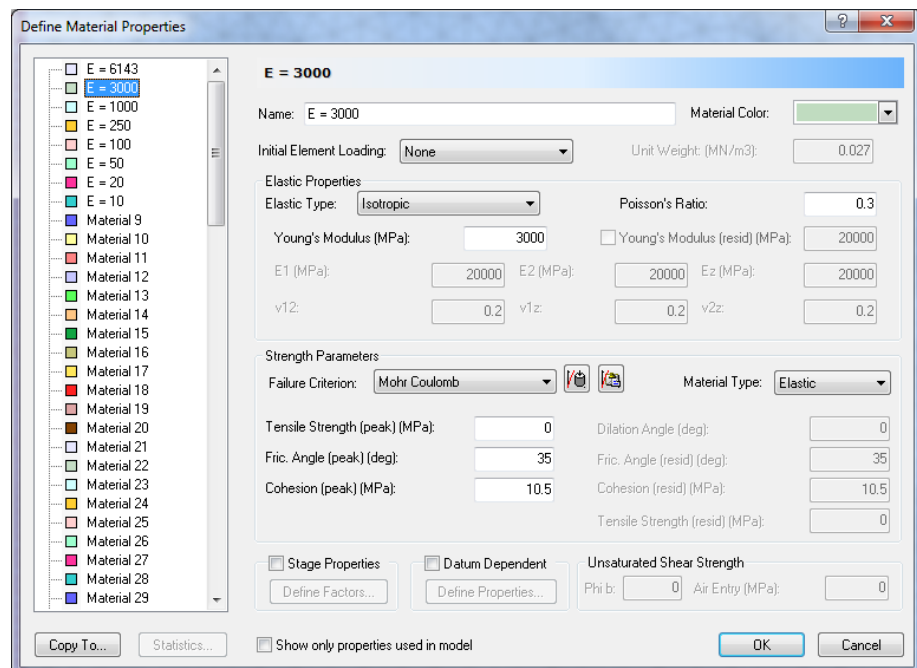
Enter Sigma 1 = 30, Sigma 3 = 15, Sigma Z = 25, Angle = 0, and select OK.





Change the Name of Material 1 to **E=6143**.

Click on the Material 2 tab and change the name to **E=3000**. Change the Initial Element Loading to None. Change the Young's Modulus to 3000 MPa. See below.



Now follow the same procedure and set the Young's modulus of Materials 3 thru 8 to 1000, 250, 100, 50, 20 and 10 MPa respectively. Change the names to reflect the value of the modulus. Make sure that the Initial Element Loading for Materials 3 thru 8 is set to None.

Click OK when done.

Now a little explanation as to what we did. The first material, with modulus 6143 MPa, and Generalized Hoek-Brown failure criterion, is the in-situ rock mass. Materials 2 thru 8 will be used inside the excavation (excavation core). The core material is progressively replaced over a number of stages. This replacement, along with the modulus reduction, allows the boundary to progressively deform. In each of the eight stages, the material inside the excavation is replaced by a material with zero internal stress (i.e. Initial Element Loading = None) and with a lower modulus than the proceeding stage. In the final stage, the material inside the excavation is removed. This process models the advancement of the tunnel face. Each stage (and corresponding core modulus) represents some distance from the tunnel face, either in front of or behind the face. The final excavated stage represents the deformed state far away from the tunnel face, at a distance where the face has no influence on stresses or displacements. What's left is determining the correspondence between core modulus and distance from the tunnel face. In particular, the modulus sequence that yields the deformation at the support installation distance. The support installation distance being the distance between the tunnel face and where the support is installed.

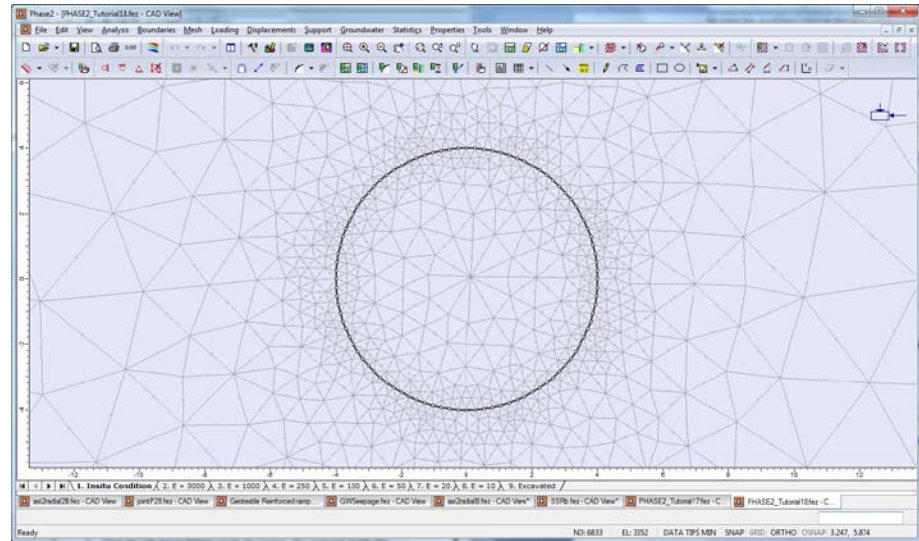
To determine the correspondence between core modulus and distance from the tunnel face, one must first know the relationship between tunnel wall deformation and distance from the tunnel face. As mentioned previously, there are a number of methods for doing this.

Knowing the relationship between tunnel wall displacement and distance from the tunnel face, and knowing the relationship between core modulus and tunnel wall displacement, you can then determine the relationship between core modulus and distance from the tunnel face. Knowing this relationship allows you to determine the modulus reduction sequence that gives the tunnel wall displacement prior to support installation.

## Core Replacement Technique

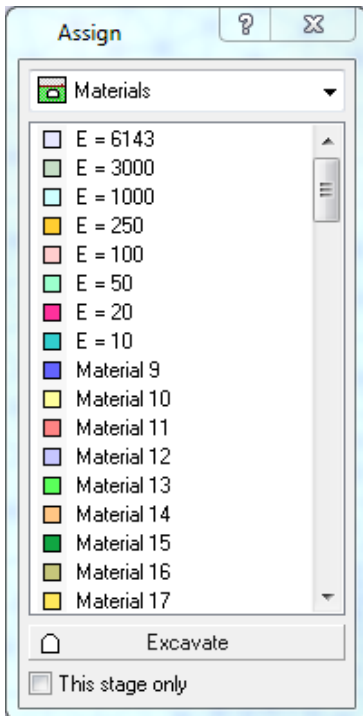


Click the Zoom Excavation button on the toolbar. You should see the following:



Select: Properties → Assign Properties

1. Make sure the Stage 2 tab, E=3000, is selected (at the bottom left of the view).
2. Select the “E=3000” button in the Assign dialog.
3. Click the left mouse button inside the tunnel. The material inside the tunnel should change to green, the color representing the E=3000 material.
4. Change to Stage 3, E=1000, by clicking the stage tab at the bottom of the screen.
5. Select the “E=1000” button in the Assign dialog.
6. Click the left mouse button inside the tunnel. The material inside the tunnel should change to light blue, the color representing the E=1000 material.
7. Change to Stage 4, E=250.
8. Select the “E=250” button in the Assign dialog.
9. Click the left mouse button inside the tunnel.
10. Change to Stage 5, E=100.
11. Select the “E=100” button in the Assign dialog.

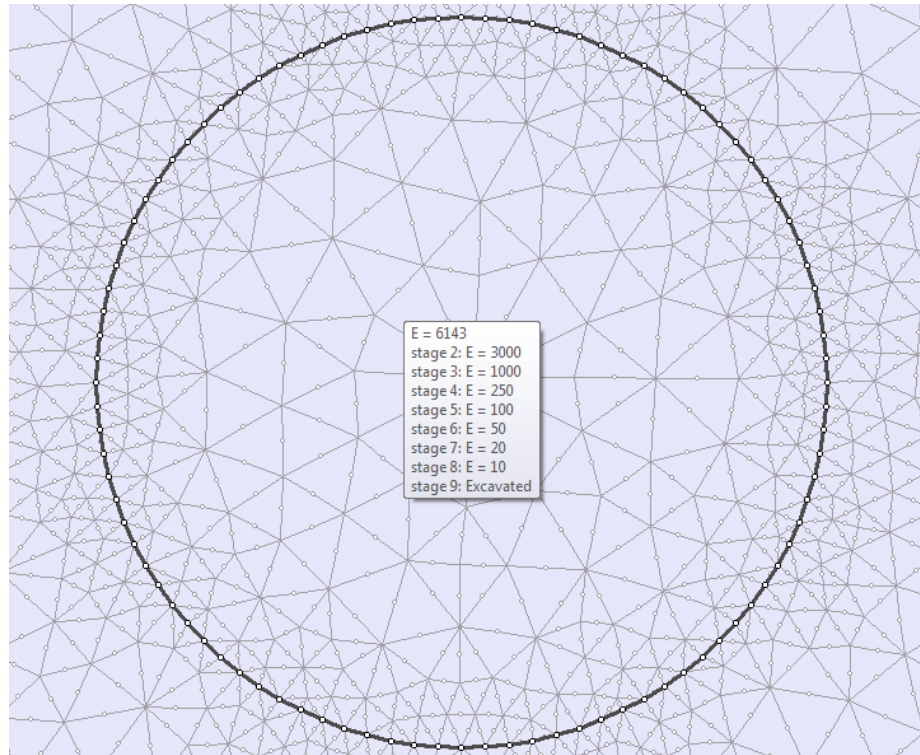


12. Click the left mouse button inside the tunnel.
13. Change to Stage 6, E=50.
14. Select the “E=50” button in the Assign dialog.
15. Click the left mouse button inside the tunnel.
16. Change to Stage 7, E=25.
17. Select the “E=25” button in the Assign dialog.
18. Click the left mouse button inside the tunnel.
19. Change to Stage 8, E=10.
20. Select the “E=10” button in the Assign dialog.
21. Click the left mouse button inside the tunnel.
22. Change to Stage 9, Excavated.
23. Select the “Excavate” button at the bottom of the Assign dialog.
24. Click the left mouse button inside the tunnel. The material inside the excavation should now be removed.
25. Close the Assign dialog by clicking on the X in the upper right corner of the dialog.

Now select stage 1 – the in-situ condition stage. Turn on the minimum data tips mode using the following command.

Select: View → Data Tips → Minimum

Hover the mouse inside the excavation. After a second, a data tip should appear. You should see the following:



Notice that the data tip shows all the materials inside the excavation as a function of stage.

We are now ready to run the analysis.

## Compute

Before you analyze your model, let's save this as a new file called **CoreSoftening.fez**



Select: File → Save

*Save the file as CoreSoftening.fez.*



Select: Analysis → Compute

The *Phase2* Compute engine will proceed in running the analysis. When completed, you will be ready to view the results in Interpret.

## Interpret

From Model, switch to the Interpret program.



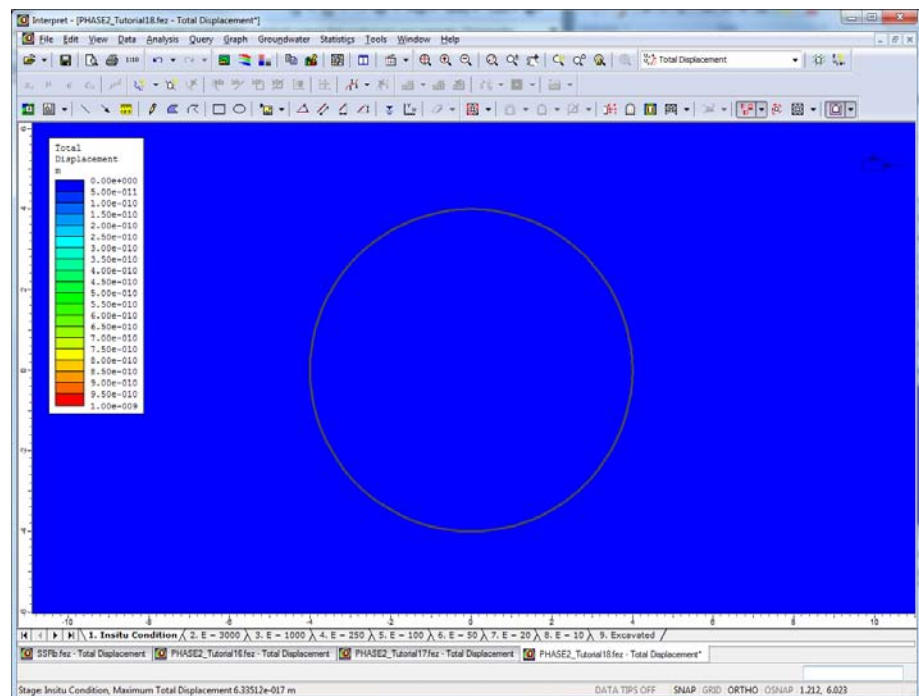
Select: Analysis → Interpret

After you select the Interpret option, the Interpret program starts and reads the results of the analysis. You will see the maximum stress, sigma 1 for Stage 1. Notice that there is no variation of stress and that the stress (30 MPa) is equal to the major in-situ field stress. This is expected since in the first stage the material inside and outside the tunnel boundary is the in-situ E=6143 material.



Now click the Zoom Excavation button on the toolbar.

Change the contours to plot Total Displacement using the pull down menu in the toolbar. The model for Stage 1 will look like this:



You can see that there no displacement in the first stage.

Now click through the stages. You'll see an increase in deformation around the tunnel as the core material is replaced and softened (modulus reduced).

## Step 1 – Computing tunnel deformation before support installation using the Vlachopoulos and Diederichs method

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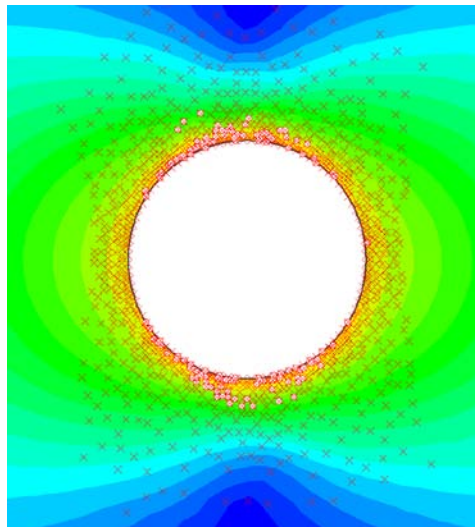
To compute the tunnel deformation at the point of support installation, we'll use the empirical relationship developed by Vlachopoulos and Diederichs. To use the Vlachopoulos and Diederichs method, you need two pieces of information from the finite-element analysis. You need to know a) the maximum tunnel wall displacement far from the tunnel face, and b) the radius of the plastic zone far from the tunnel face.

Both of these values can be computed from a plane strain analysis with zero internal pressure inside the excavation. In the model we just built, the results from stage 9 are used since the material inside the excavation is completely removed in this stage.

Switch to the last stage, stage 9. Look in the lower left corner of the program window on the status bar. You'll see that the maximum displacement for this stage is approximately 0.061m. This is the value of maximum wall displacement far from the tunnel face. The location of this displacement is in the roof and floor of the excavation. The location of this displacement is important since any comparisons of displacement for various core moduli must be made at the same location.



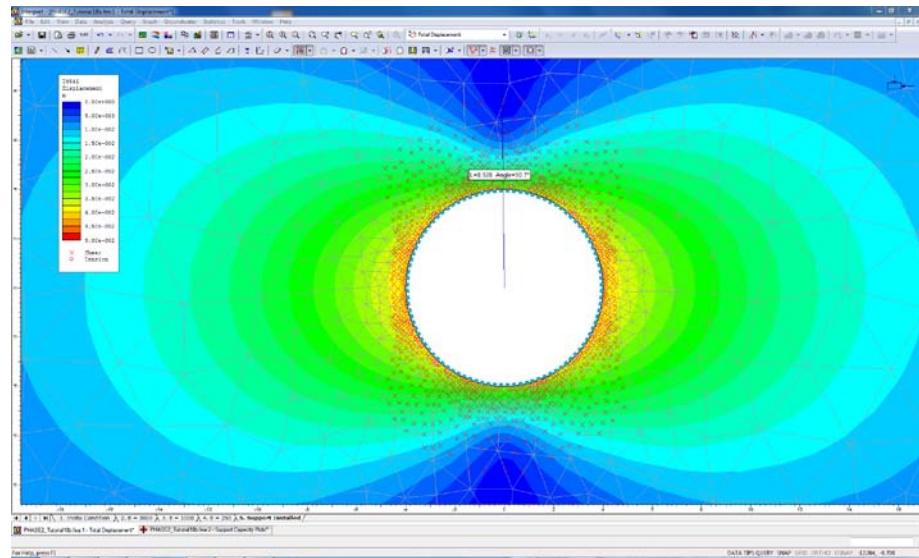
To determine the radius of the plastic zone, first turn on the display of yielded elements using the Display Yielded Elements  toolbar button. You'll see a number of crosses representing elements in the finite element analysis that have failed. **Zoom Out**  so that the entire extent of failed points is visible (see below).



The extent of this failed zone represents the extent of the plastic zone around the tunnel. To determine the radius of the plastic zone, you can use either the measuring tool or the dimensioning tool to measure the distance from the center of the tunnel to the perimeter of the yielded/plastic zone. In this tutorial we'll use the measuring tool.

Select: Tools → Add Tool → Measure

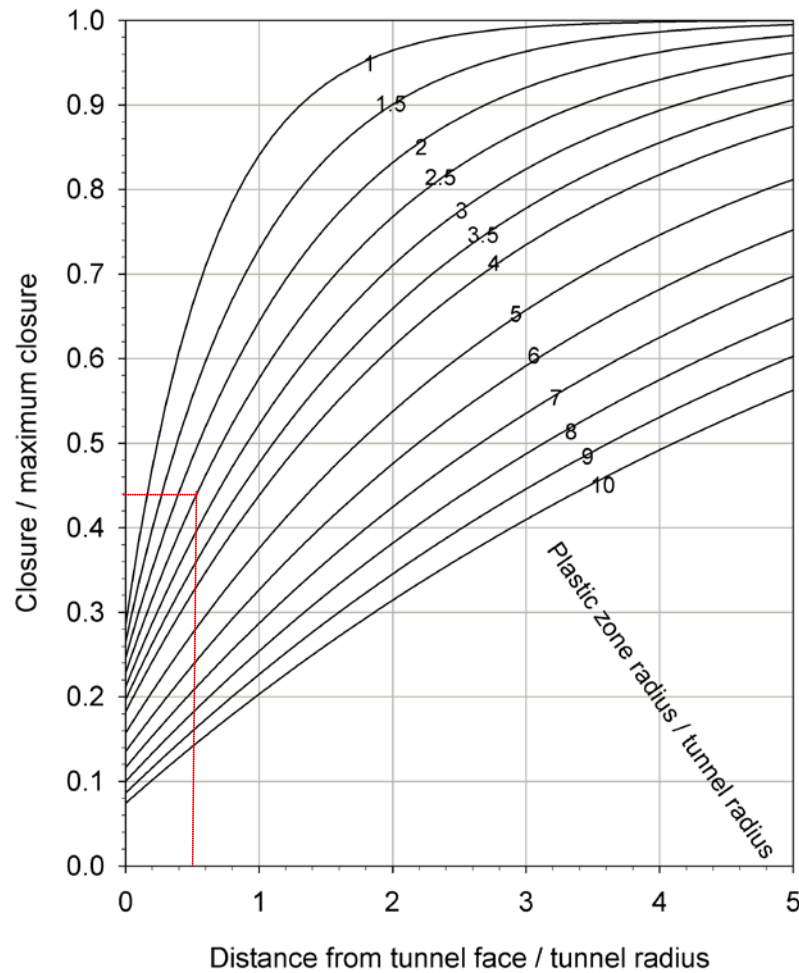
Pick the location to measure from [esc=quit]: 0,0  
 Pick the location to measure to [esc=quit]: ***use the mouse to extend the measuring line vertically until you get to the edge of the yield zone, press the left mouse button.***



As seen above, the radius of the plastic zone is approximately 9.5 m.

#### Computing displacement prior to support installation using the Vlachopoulos and Diederichs Method

The following plot was created using the Vlachopoulos and Diederichs equations (Vlachopoulos and Diederichs, 2009). The equations can also be found in the Kersten Lecture, appendix 1 (Hoek et. al., 2008). Using this plot, you can estimate the amount of closure prior to support installation if you know the plastic radius and displacement far from the tunnel face.



For our problem,  $R_p=9.5\text{m}$ ,  $R_t=4\text{m}$ ,  $X=2\text{m}$ , and  $u_{\max}=0.061\text{m}$ . The Distance from tunnel face/tunnel radius =  $2/4 = 0.5$ . The Plastic zone radius/tunnel radius =  $9.5/4 = 2.4$ . From the above plot this gives Closure/max closure approximately equal to 0.44. Therefore the closure equals  $(0.44) \cdot (0.061) = 0.027\text{m}$ .

As computed above, the tunnel roof displaces 0.027m before the support is installed.

## Step 2 - Determining the core modulus

The next step is to determine the core modulus that yields a displacement of 0.027m in the roof of the tunnel. It is important to maintain the same location as is used to determine  $u_{\max}$ , since the location of maximum displacement can change depending on the magnitude of the internal pressure. This can be seen in this model as larger core moduli produce larger displacement in the sidewall while smaller core moduli produce larger displacements in the roof and floor.

To determine the internal pressure that yields a 0.027m roof displacement, we'll plot the displacement versus stage for a point on the roof of the excavation.

Make sure you have Total Displacement selected as the data type.

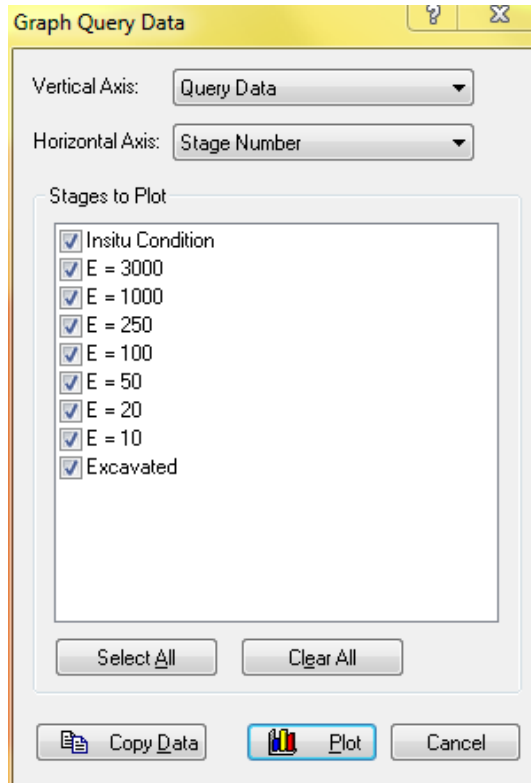
### Graphing Displacement in the Roof of the Excavation

To create the graph:

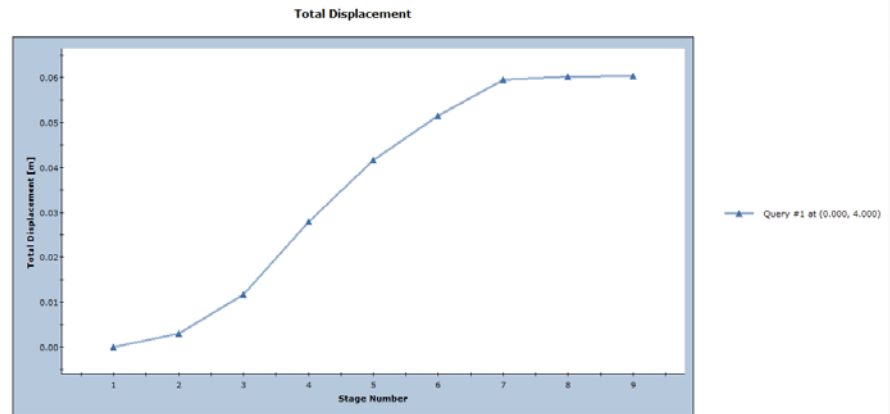


Select: Graph → Graph Single Point vs. Stage

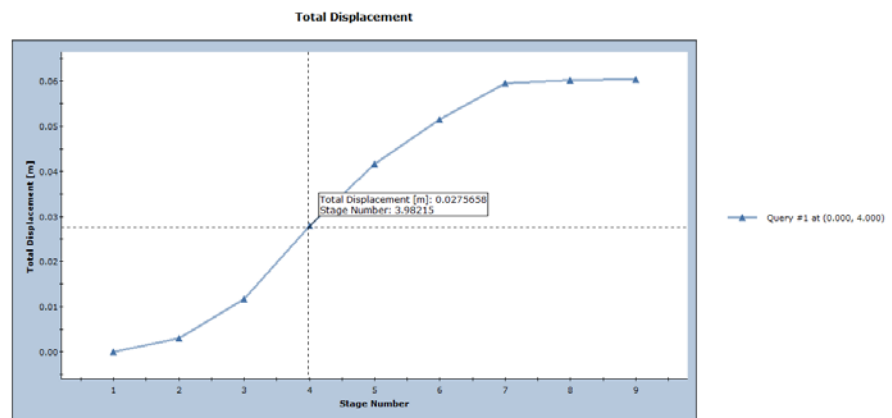
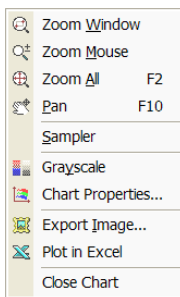
1. When asked to enter a vertex, type in the value 0,4 for the location and press Enter. This is a point on the roof of the excavation.
2. You will see the Graph Query Data dialog.



3. Press the Plot button. The following figure shows the plot generated by the program. This is a plot of displacement versus stage for a point in the roof of the tunnel.



Right-click in the plot and choose the **Sampler** option. Move the sampler by moving the mouse with the left mouse button. Move the sampler until the displacement value on the right side of the plot is equal to 0.027m.



From this plot, you can see that in stage 4, the wall displacement in the roof of the tunnel is approximately 0.027m. This represents a 3 stage material replacement and reduction of core modulus from  $E=6143$ (insitu), to 3000, 1000 and finally 250 MPa.

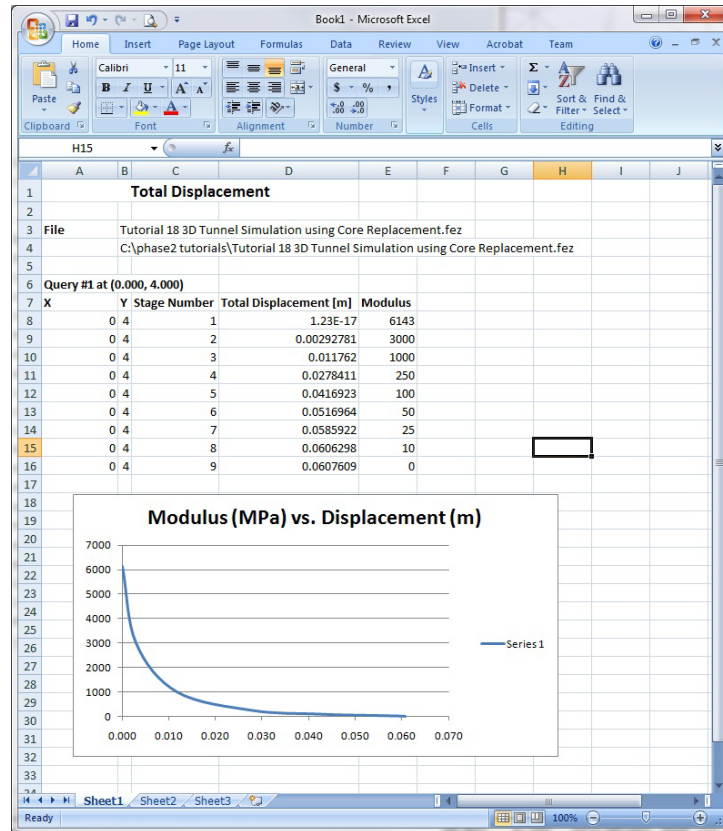
### Creating a convergence confinement graph in Excel

Often you want to create a convergence confinement graph which plots displacement versus core modulus. This is easily done by exporting the above graph to Microsoft Excel™. This requires that you have Excel installed on your computer.

Right-click in the Graph you just created and choose the **Plot in Excel** option.

Excel will launch with a plot of stage number versus displacement. You can easily modify the plot to change the stage number data to the core modulus. A sample of the Excel file for this example is included in the Tutorials folder with the *Phase2* data files.

The following image shows the convergence-confinement plot in Excel for this example. You can see by this plot that modulus reduction to 250MPa yields the tunnel wall displacement computed above for the point of support installation (0.027m).



We have now completed steps 1 and 2 as defined in the Problem section at the beginning of this tutorial. It is now time to actually design our support system.



From Interpret, switch back to the *Phase2* Model program by pressing the **Model** button on the toolbar.

**IMPORTANT:** see the note at the end of this tutorial, about how to carry out the analysis if the required modulus value lies between two values in your initial modulus reduction sequence.

## Model

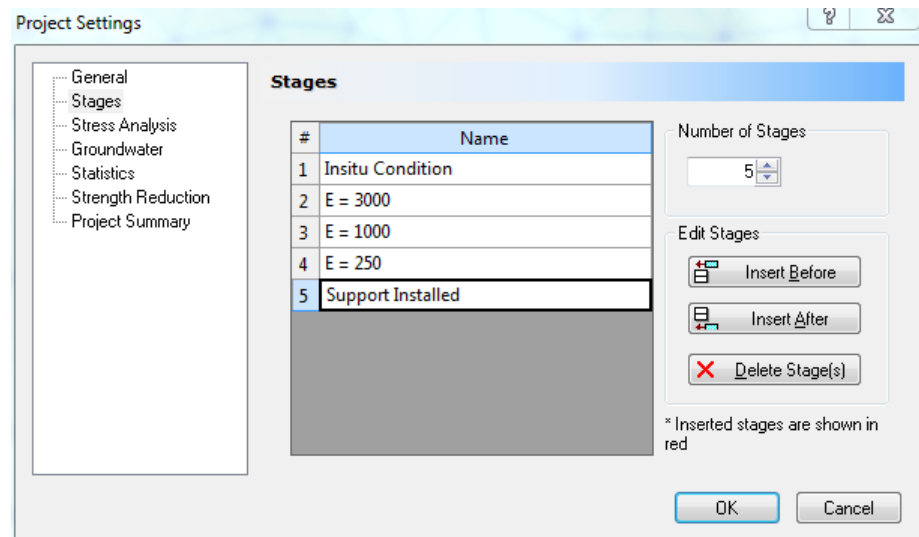
You should now be in the *Phase2* Model program with the 9 stage model you created above loaded into the program.

We will use this file and modify it to do the support design.

## Project Settings



Open the **Project Settings** dialog from the **Analysis** menu and select the Stages Tab. Delete stages 5,6,7, and 8. Note: you can select multiple stages by scrolling down the number column with the left mouse button depressed. Use the Delete Stages button to delete the stages. Change the name of stage 5 from *Excavated* to *Support Installed*. The dialog should look like:



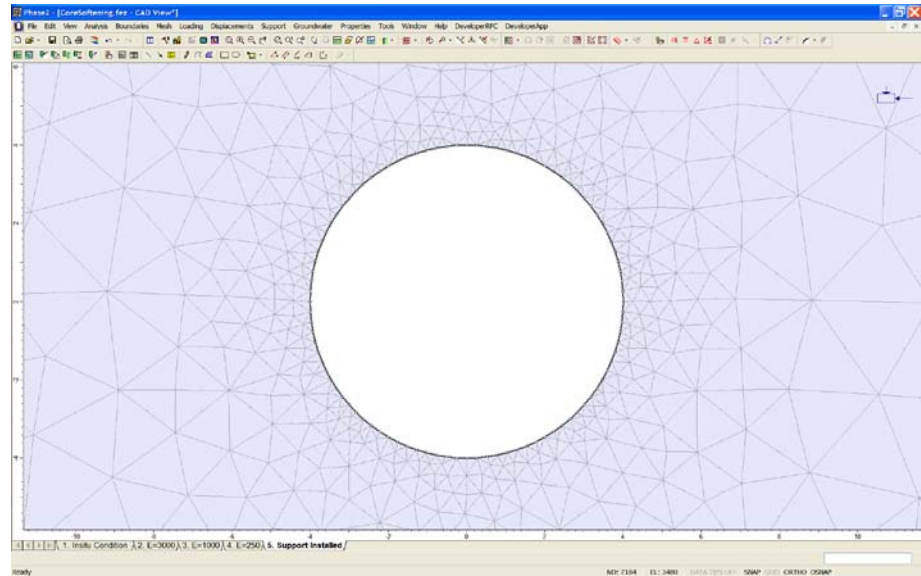
It is important that we keep all the core softening stages up to the stage that represents support installation. This is because the replacement and softening of the core material in stages 2 and 3 affect the final displacement result. These stages directly influence the stress path and displacement of the material around the excavation.

Close the dialog by clicking OK.



Make sure the Stage 5, Support Installed stage tab is selected. Click the Zoom Excavation button on the toolbar.

You should see the following:



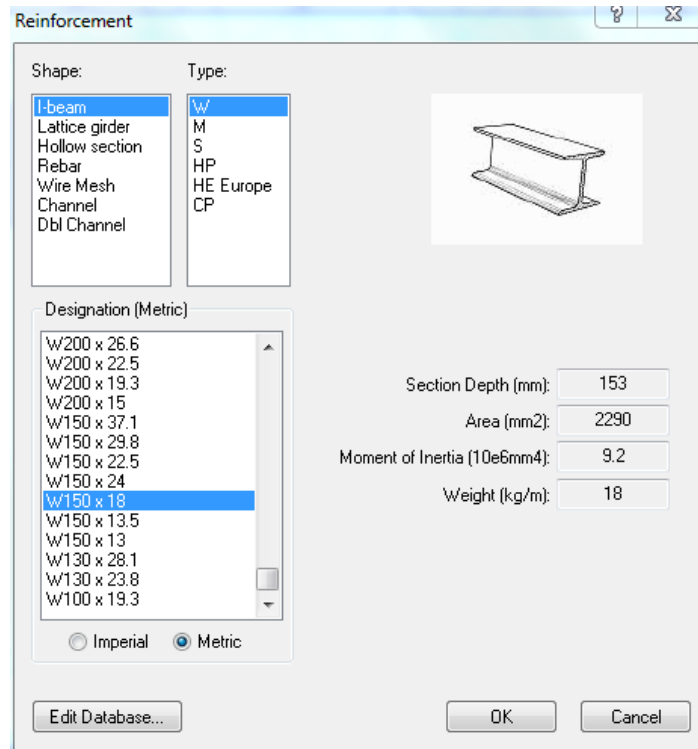
## Setting the Reinforced Concrete Liner Properties

Now define the liner properties. The properties we enter will correspond to a 200 mm thick layer of concrete reinforced with W150X18 I-beams spaced at 2 meter intervals along the tunnel axis.

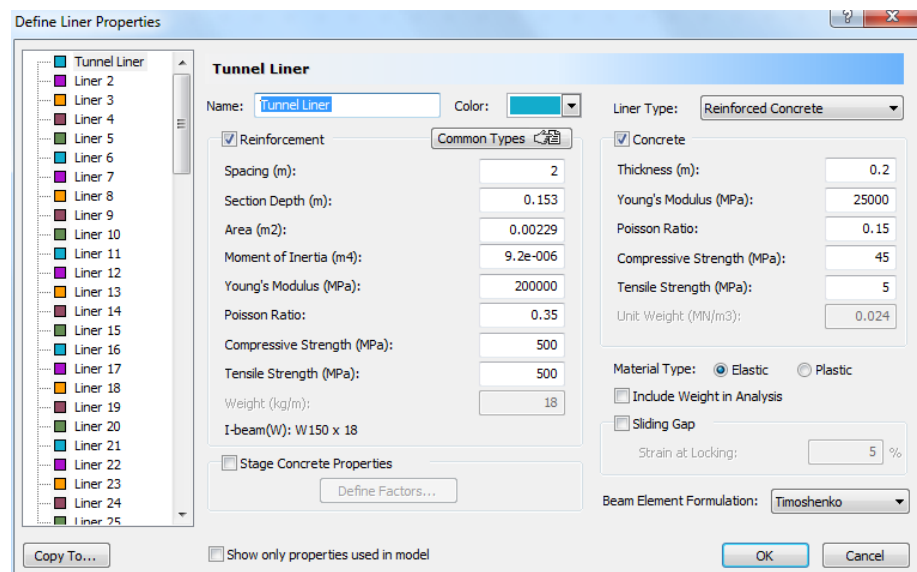


Select: Properties → Define Liners

1. Change the Name of the liner to Tunnel Liner
2. Change the Liner Type to **Reinforced Concrete**
3. Click on the **Common Types** button. You will see the Reinforcement database dialog shown below. For the Reinforcement, we will select an I-beam from a list of standard reinforcement types.
4. In the Reinforcement database dialog, select the W150 x 18 I-beam. Click OK, and the I-beam reinforcement properties will be automatically loaded into the Define Liner Properties dialog.



5. In the Define Liner Properties dialog, for the Reinforcement, enter a spacing of 2m.
6. Enter the properties for the concrete. Thickness=0.2m, Modulus=25000MPa, Poisson Ratio=0.15, Compressive Strength=45MPa, Tensile Strength=5MPa. The liner properties dialog should look like:



7. Press OK to save your input and exit the dialog.

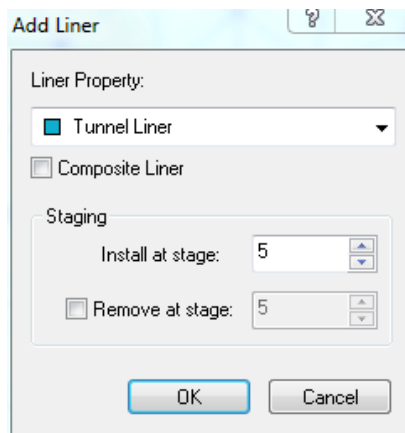
## Adding a Reinforced Concrete Liner to the Tunnel

We will now line the tunnel with the liner defined above. First make sure that Stage 5, the Support Installed stage, is selected.

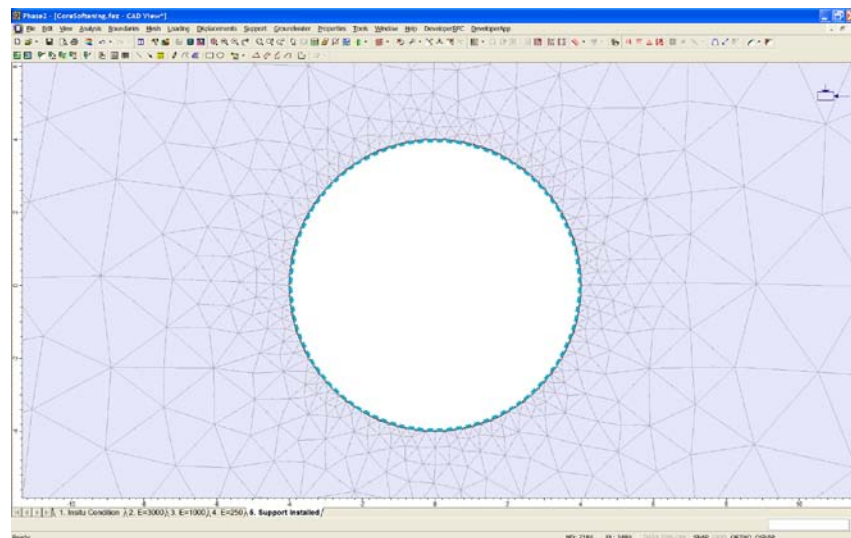


Select: Support → Add Liner

1. You will see the Add Liner dialog. Make sure it looks like the following image. Select OK.



2. Click and hold the left mouse button, and drag a selection window which encloses the entire excavation. Release the left mouse button. Notice that all excavation line segments are selected.
3. Right-click the mouse and select Done Selection, or just press the Enter key. The entire tunnel will now be lined, as indicated by the thick blue line segments around the excavation boundary (see below).



Click through the stages. Notice how the color of the liner changes from light blue in stages 1 thru 4 to dark blue in stage 5. This indicates that the liner is being installed in stage 5.

We are now ready to run the analysis.

## Compute

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Before you analyze your model, let's save this as a new file called **CoreSofteningLinerDesign.fez**. (Make sure you select Save As and not Save, or you will overwrite the internal pressure reduction file).

Select: File → Save As

*Save the file as CoreSofteningLinerDesign.fez.*



Select: Analysis → Compute

The *Phase2* Compute engine will proceed in running the analysis. When completed, you will be ready to view the results in Interpret.

## Interpret

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From Model, switch to the Interpret program.



Select: Analysis → Interpret

If any other files are loaded in the Interpret program (i.e. the *CoreSoftening.fez* file), close them. Click on the tab at the bottom of the program window associated with the file and use the File→Close menu option to close the file.



Make sure the Stage 5 tab is selected. Click the Zoom Excavation button on the toolbar.

## Support Capacity Diagrams

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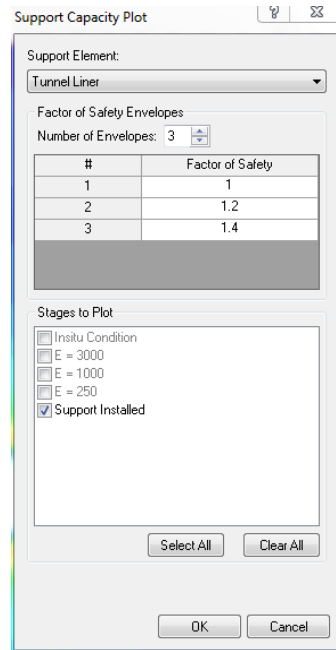
Support capacity diagrams give the engineer a method for determining the factor of safety of a reinforced concrete liner. For a given factor of safety, capacity envelopes are plotted in axial force versus moment space and axial force versus shear force space. Values of axial force, moment and shear force for the liner are then compared to the capacity envelopes. If the computed liner values fall inside an envelope, they have a factor of safety **greater than** the envelope value. So if all the computed liner values fall inside the design factor of safety capacity envelope, the factor of safety of the liner exceeds the design factor of safety.



Select: Graph → Support Capacity Plots

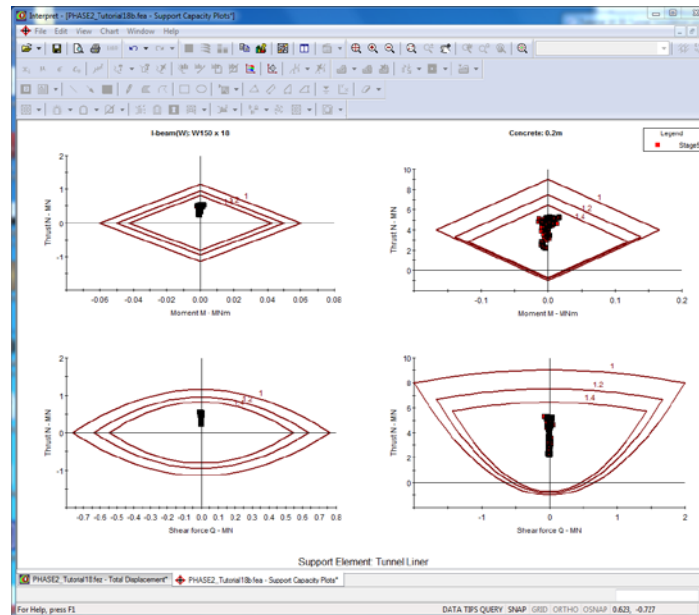
The Support Capacity Plot dialog allows you to choose the support element (i.e. liner type), the number of envelopes, and the stages from which the liner data is taken.

Use the spin control to increase the number of envelopes to 3. The dialog should look like:



Press OK.

The following plot is generated. The dark red lines represent the capacity envelopes for the 3 factors of safety (1, 1.2, 1.4).



Right away you'll notice that all the data points fall within the factor of safety=1.4 envelope, on all four plots. This means that the support system chosen has a factor of safety greater than 1.4.

For further information on some of the tools that can be used with support capacity plots, see tutorial 24.

## Note about determining the final core modulus

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Before we conclude this tutorial, it is important to note the following.

In this example, the required core modulus which gives the displacement required at the point of support installation, just happens to be exactly equal to one of the original modulus values chosen for the initial reduction sequence (i.e. 250 MPa). In general this will not be the case. That is, the required core modulus will probably lie between two of the values chosen for your initial modulus reduction sequence. If this occurs, you should do the following:

1. Use the convergence-confinement graph to determine the required core modulus at the point of support installation, as discussed earlier in this tutorial.
2. Then you can either insert a new stage of core replacement, with the required modulus value, or simply use the nearest stage with a HIGHER modulus value, and lower the material modulus at this stage to the required value (e.g. if the required modulus is 350 MPa, but your initial sequence goes from 500 to 250, then change the 500 value to 350).
3. Re-run the analysis and check if the new modulus value does in fact give the desired displacement at the point of support installation. It should be close. If not, then repeat steps 1 to 3 until you determine the required modulus value.

This concludes the tutorial; you may now exit the *Phase2* Interpret and *Phase2* Model programs.

## References

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